

Dongjun Kim

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RESEARCH INTERESTS

Energy-Efficient AI Systems

Algorithm-Hardware Co-Design, Efficient AI Computing, Brain-Inspired Neuromorphic Computing

EDUCATION

Kangwon National University

B.S. Electronic and Semiconductor Engineering (GPA: 4.28/4.5, Rank: 2/67)

Military Service, Republic of Korea Army(Sergeant)

Gangneung, South Korea

March 2021 – Feb 2027

Jan 2022 – July 2023

PUBLICATIONS

- **Quantization Adversarial Distillation for Embedded Underwater Acoustic Target Recognition**

By **Dongjun Kim**, Sung-Hoon Byun, Sangwook Park*

Published at *IEEE Embedded Systems Letters (ESL)*, 2026 [[Paper](#)] [[Code](#)]

- **Ultra-Lightweight Ship-Radiated Sound Classification for Real-time Embedded Inference**

By Sangwon Park, **Dongjun Kim**, Sung-Hoon Byun, Sangwook Park*

Published at *IEEE Embedded Systems Letters (ESL)*, 2026 [[Paper](#)] [[Code](#)]

RESEARCH EXPERIENCE

Kangwon National University

Undergraduate Researcher

Feb 2025 – Present

Gangneung, South Korea

- Advised by **Prof. Sangwook Park** at **Laboratory for Machine Audio Perception(L-MAP)**
- Developed QAD (Knowledge Distillation + Adversarial Training + QAT) for underwater acoustic target recognition on resource-constrained edge devices. Achieved **74.11% accuracy** with **63% model size reduction** and **56% inference latency reduction** on **Raspberry Pi 5**.
- Investigating Sonar-Equation-Based Physics-Informed Diffusion Models to Overcome Data Scarcity in Underwater Acoustic Target Recognition.

Purdue University

Visiting Undergraduate Researcher

July 2026 – August 2026

West Lafayette, USA

- Advised by **Prof. Martin Byung-Guk Jun** and **Dr. Asif Tanveer**.
- Developed and deployed a real-time Edge AI system for **CNC machine chatter detection** in a real-world manufacturing environment. [[Slides](#)]
- Received **Outstanding Individual Award** and **Outstanding Project Team Award** (ranked 1st among 21 participants and 7 teams, respectively).

CONFERENCE PRESENTATIONS

- **A Study on Physics-Informed Diffusion for Acoustic Data Augmentation**

By, **Dongjun Kim**, Sung-Hoon Byun, Sangwook Park*

Submitted at *IEEE/IEIE International Conference on Consumer Electronics Asia(ICCE-Asia)*, 2026

- **Adversarial Quantization based Knowledge Distillation for low-complexity vessel sound classification**

By, **Dongjun Kim**, Sung-Hoon Byun, Sangwook Park*

Presented at *The Acoustical Society of Korea (ASK)*, 2026

- **An Approach for Minimizing Processing Error in Sound Event Detection based on BEATs-BiGRU model**
By Dongjun Kim, Sangwook Park*
Presented at The Institute of Electronics and Information Engineers (IEIE), 2025

PROJECTS

On-ieu: Integrated Public Transportation AIoT System for the Digitally Vulnerable *Mar 2026 – Present*

- **Role:** Technical Lead (Capstone Project) — Designed end-to-end system pipeline.
- Developed an end-to-end navigation assistance system for digitally vulnerable populations without smart-phones, integrating face detection, speech-to-text (STT), an On-device LLM, and public transit APIs.
- Deployed STT, Face Recognition, and Large Language Models on **Edge Devices (Raspberry Pi 5 with AI HAT+, NVIDIA Jetson Orin Nano)** for real-time service. [\[Code\]](#)

TECHNICAL SKILLS

- **Programming Languages:** Python, C
- **Frameworks & OS:** PyTorch, Linux, Git, ONNX
- **Hardware Platforms:** Raspberry Pi, NVIDIA Jetson Nano/Orin Nano

HONORS & AWARDS

Full-Tuition Merit Scholarship, Kangwon National University	September, 2026
Outstanding Project Team & Outstanding Individual, Purdue-KIAT AI-track Program	August 2026
System-Semiconductor Micro Degree Scholarship, Kangwon National University	March 2026
Outstanding Paper Award, IEIE Conference on Electronics, Semiconductor & AI	August 2025
Academic Excellence Scholarship, Kangwon National University	2024 – 2025
Full-Tuition Merit Scholarship, Kangwon National University	Fall 2021

TEACHING EXPERIENCE

Teaching Assistant, Advanced Programming (C language)	Spring 2026
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